

Indian Institute of Technology Guwahati
Statistical Inference and Multivariate Analysis (MA324)
Problem Set : Delta Method and Bootstrap

1. Let $X_1, X_2, \dots, X_n$ be an i.i.d. sample from a $N(\mu, \sigma^2)$ distribution. Define the transformation $g(\bar{X}_n) = (\bar{X}_n - \mu)^2$.
 - (a) Demonstrate why the standard first-order Delta Method fails to find the asymptotic distribution of $\sqrt{n}(g(\bar{X}_n) - g(\mu))$.
 - (b) Derive the correct asymptotic distribution using the second-order Taylor expansion and state the appropriate normalizing sequence (rate of convergence).
2. Let $X_1, X_2, \dots, X_n$ be an i.i.d. random sample from a distribution where the mean is μ and the variance is $\sigma^2(\mu) = c\mu^2$ for some known constant $c > 0$.
 - (a) Derive a variance-stabilizing transformation $g(\bar{X}_n)$ such that the asymptotic variance of $g(\bar{X}_n)$ does not depend on μ .
 - (b) Prove the asymptotic normality of your transformed statistic.
3. Let $X_1, \dots, X_n$ be i.i.d. $Uniform(0, \theta)$. It is a known result that the maximum order statistic $X_{(n)}$ converges in distribution to a non-degenerate limit when scaled by n : $n(\theta - X_{(n)}) \xrightarrow{d} Exp(1/\theta)$.
 - (a) Prove analytically why the standard non-parametric bootstrap fails to provide a consistent estimator for the sampling distribution of $X_{(n)}$.
 - (b) Read (online) about m -out-of- n bootstrap as an alternative resampling strategy that resolves this inconsistency.
4. Let $X_1, X_2, \dots, X_n$ be an i.i.d. random sample from a population modeled by a two-component Weibull mixture distribution. The probability density function is given by

$$f(x; \boldsymbol{\theta}) = p \cdot w(x; \alpha_1, \gamma_1) + (1 - p) \cdot w(x; \alpha_2, \gamma_2),$$

where the Weibull density is parameterized by shape $\alpha > 0$ and rate $\gamma > 0$ as:

$$w(x; \alpha, \gamma) = \alpha \gamma x^{\alpha-1} \exp(-\gamma x^\alpha) \quad \text{for } x \geq 0$$

The parameter vector $\boldsymbol{\theta} = (\alpha_1, \gamma_1, \alpha_2, \gamma_2, p)$ is estimated via maximum likelihood to obtain $\hat{\boldsymbol{\theta}}$. Write down the step-by-step algorithmic procedure (pseudo code) to construct a 95% parametric (and non-parametric) bootstrap confidence interval for the mixture weight parameter. Then, implement the procedure using a language of your choice.